

NOVO SELO, Bulgaria

WMO: 155010

Lat: 44.160N

Lon: 22.785E

Elev: 125

StdP: 14.63

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	15.8	6.9	7.9	11.3	13.1	10.8	16.8	32.3	41.8	27.7	39.0	2.9	50	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	92.9	74.3	90.2	73.8	87.5	72.6	76.5	89.3	75.1	88.0	73.5	85.6	5.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	119.9	85.3	70.8	114.0	83.6	69.3	108.2	81.4	40.0	89.4	38.6	88.3	37.1	85.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	23.0	20.2	5.1	96.7	8.4	4.1	-0.9	99.7	-5.8	102.1	-10.5	104.4	-16.6	107.4		
(5)			4.8	79.8	8.6	2.2	-1.4	81.4	-6.4	82.7	-11.2	83.9	-17.5	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	32.3	34.9	44.6	55.3	64.7	71.3	75.8	74.6	66.1	54.1	43.5	34.5
	DBStd	16.91	9.11	9.64	8.22	5.37	5.81	5.72	4.69	4.61	5.83	7.07	6.96	7.21
	HDD50	1929	549	425	202	15	0	0	0	0	0	40	216	482
	HDD65	4869	1012	843	632	294	81	13	1	3	57	344	644	945
	CDD50	3541	1	2	35	174	454	640	798	764	482	167	22	2
	CDD65	1007	0	0	0	3	70	203	334	302	89	6	0	0
	CDH74	9511	0	0	2	35	562	1927	3414	2811	713	47	0	0
	CDH80	3638	0	0	0	3	98	698	1499	1146	191	3	0	0
	WSAvg	5.6	5.1	5.8	7.2	5.6	5.9	5.7	5.6	5.5	5.5	4.9	4.9	5.2
	PrecAvg	21.9	1.5	1.6	1.3	1.9	2.2	2.1	2.0	1.7	1.7	1.8	1.5	2.4
(15) Precipitation	PrecMax	31.3	2.5	3.9	2.8	3.0	4.7	6.5	4.7	6.1	4.1	6.1	3.8	5.5
	PrecMin	12.3	0.2	0.0	0.2	0.0	1.0	0.5	0.3	0.0	0.1	0.1	0.0	0.8
	PrecStd	4.9	0.7	1.1	0.9	0.8	1.0	1.5	1.3	1.6	1.3	1.6	1.1	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	60.4	71.2	78.8	86.1	93.2	99.6	96.0	89.1	79.8	65.7	57.5
		MCWB	48.3	50.6	54.4	63.3	69.3	75.1	74.9	74.5	70.0	65.7	56.7	50.1
	2%	DB	51.2	55.7	65.6	72.6	82.8	90.0	94.2	92.2	85.1	73.4	60.3	49.6
		MCWB	45.4	48.6	53.8	59.1	68.7	74.4	74.4	74.5	69.3	63.3	54.3	45.8
	5%	DB	47.0	51.2	61.0	69.1	80.1	87.0	90.6	89.5	81.7	69.0	56.6	46.0
		MCWB	42.7	45.8	51.5	57.6	67.4	72.9	74.0	73.2	67.6	61.2	52.6	43.0
	10%	DB	43.1	47.0	56.9	66.1	77.2	84.0	87.7	86.5	78.0	65.1	53.6	43.3
		MCWB	40.3	43.2	49.2	56.3	66.0	71.3	73.1	71.9	65.9	59.0	50.7	41.2
	0.4%	WB	49.7	51.8	56.6	64.6	71.9	77.5	77.9	78.2	72.2	67.6	58.2	50.6
		MCDB	56.3	58.2	67.8	76.7	81.9	89.7	91.6	90.2	84.0	76.2	63.9	56.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.4	48.7	54.1	60.4	69.8	75.6	76.4	76.3	70.1	64.5	55.7	46.8
		MCDB	50.7	54.9	64.2	69.5	80.8	87.5	89.8	88.7	83.1	71.3	58.9	49.2
	5%	WB	43.0	46.0	52.0	58.7	68.2	73.7	75.2	74.5	68.4	61.8	53.2	43.7
		MCDB	46.5	50.5	60.2	67.1	78.8	85.2	88.9	87.1	80.1	68.4	56.0	45.4
	10%	WB	40.8	43.4	49.6	56.8	66.6	72.0	73.7	72.8	66.7	59.2	50.9	41.6
		MCDB	43.0	46.8	56.2	64.9	76.1	83.1	86.0	84.7	76.5	64.7	53.5	43.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	11.0	15.1	17.1	18.5	20.0	21.3	20.7	18.8	14.8	10.7	8.4
		MCDBR	15.8	18.6	22.4	23.2	23.5	24.8	26.3	25.4	25.0	21.0	15.7	12.7
		MCWBR	10.7	12.5	13.1	12.8	12.1	11.8	12.2	12.1	13.0	12.8	10.3	9.4
		MCDBR	14.4	18.1	21.2	20.5	22.1	23.3	24.6	23.5	23.1	19.3	13.6	10.7
	5% WB	MCWBR	10.1	12.3	13.1	12.4	12.3	11.5	12.0	12.3	12.6	12.4	9.3	8.5
(40) Clear-Sky Solar Irradiance	taub		0.346	0.374	0.412	0.472	0.453	0.446	0.459	0.476	0.445	0.404	0.373	0.346
	taud		2.283	2.199	2.173	2.026	2.148	2.207	2.177	2.123	2.173	2.281	2.312	2.311
	Ebn at Noon		239	252	259	251	261	263	258	248	245	239	226	226
	Edh at Noon		27	35	41	52	47	45	45	46	41	32	27	24
(44) All-Sky Solar Radiation	RadAvg		441	697	1074	1453	1774	1998	2008	1785	1273	828	477	373
	RadStd		64	101	140	164	154	169	135	159	112	116	68	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	+1.08	N/A	N/A	N/A	+0.68	N/A	N/A	-275	+182	N/A

Nomenclature: See separate page